PRODUCT REQUIREMENTS DOCUMENT

Agentic Reality-Reconciled Digital Twin

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1. Strategic Context

Problem Statement

User	Site Superintendent on a hyperscale construction site (Data Center EPC)
Pain	Daily schedule updates rely on manual data collection — the Superintendent personally chases each subcontractor for status, translates verbal reports into P6 fields, and hunts for gaps before they become delays. Current software turns a site management role into a data entry role.
Evidence	Gaps between trades during handoffs account for an estimated 65% of major project delays. Fast GTM requirements for Data Center EPCs make schedule protection the single most critical operational lever.
Workaround	Morning walkdowns, radio check-ins, whiteboard trackers, and end-of-day manual P6 updates by the Superintendent or PM.

User Personas

Persona	Role	Motivation	Job to Be Done
Project Executive / VP Ops	Economic Buyer	Protect schedule and margin on flagship data center builds	Sign off on tools that demonstrably reduce schedule risk
Site Superintendent	Primary User	Manage execution without drowning in data entry	Know where every trade stands without asking anyone
Specialty Foreman	Secondary User	Log progress quickly and get back to work	Report status in seconds without opening an app
Project Manager	Secondary User	Accurate billing and trade coordination data	Pull schedule actuals without chasing the Super

Strategic Objectives

Objective	Target	Timeframe
Problem validation	50% positive response rate on 100 outreach attempts	Concept (30 days)

Objective	Target	Timeframe
Pilot sign-ups	15 qualified prospects → 3 pilot deployments	Prototype (30 days)
Pilot-to-production contracts	≥1 signed contract	End of pilot (60 days)
Beta commercialization	15 customers, daily use, testimonials, financial agreements	Scale (90 days)
Schedule improvement	≥30% reduction in superintendent chase time	60 days post-pilot

2. Data Specifications

Data Ingredient List

Source	Type	Volume / Window	Owner
Primavera P6	Schedule activity data	Full project schedule; refreshed per sync cycle	GC Scheduler
BIM Core Object Library	3D model objects	Full project model; updated per design revision	BIM Manager
Drone Reality Capture	Point cloud / photogrammetry	Per flight (3 flights during pilot)	Terabase / Site
Foreman SMS / Email	Unstructured natural language	Per shift; ≥1 update per active trade per day expected	Specialty Foreman

Quality Thresholds

Metric	Threshold	Consequence if Unmet
P6 freshness	Updated within current shift before agent sync	Agent suggestions flagged as stale; superintendent warned
BIM completeness	Core objects present for all active activities	Agent cannot validate drone data; conflict surfaced to Super
Foreman report rate	≥1 update per active trade per day	Activity marked as gap; async prompt sent to foreman
Drone data quality	Point cloud coverage ≥80% of scanned area	Partial scan flagged; agent uses available data with caveat

Data Supply Chain

- **P6:** GC scheduler owns and maintains. Agent reads via API or scheduled export. If P6 is unavailable, agent queues updates and retries; superintendent is notified of the delay.
- **BIM:** BIM Manager owns. Treated as ground truth for object validation. Discrepancies between BIM and field reports surface as conflicts — not agent errors.
- **Drone data:** Ingested post-flight. Agent cannot act on drone data until ingest is complete. Partial or failed ingests are flagged, not silently skipped.

- **Foreman reports:** Delivered via SMS or email. Ownership of report accuracy rests with the reporting foreman.

3. Reasoning Architecture

> **Core principle:** The agent is a preparer, not an executor. It eliminates the data entry burden and surfaces the right decisions to the right human — it does not make schedule decisions autonomously.

Authority Boundaries

Capability	Disposition	Notes
Map foreman report to schedule activity	Autonomous	100% routing accuracy required; mismatches trigger clarification flow
Estimate % complete from report	Autonomous	Surfaced with confidence score; low-confidence queued for review
Send async clarifying message to foreman	Autonomous	Non-blocking; does not hold up draft update
Flag gap (no report received)	Autonomous	Sent to superintendent review queue within 15 min
Flag source conflict (drone vs. foreman)	Autonomous	Both versions drafted; conflict reason stated; Super picks
Push AI draft update to superintendent queue	Autonomous	Pre-populated suggestion; Super reads and edits before approving
Write approved update to P6	Requires Super approval — always	No exceptions. Never automated regardless of confidence level.
Override or roll back a P6 entry	Requires Super approval — always	Super can reverse any agent-proposed update after approval

Decision Logic

When a foreman report is received:

- Map message to schedule activity. Must be near-100% confident or trigger async clarification to foreman.
- Estimate magnitude (% complete). Surface confidence score alongside estimate.

- If magnitude confidence is below threshold → queue for review and send async clarification to foreman.
- If drone data exists for same activity → compare. If sources agree, draft update. If sources conflict, draft both versions, flag the conflict with suspected reason, surface to superintendent.
- Push AI draft update to superintendent review queue within 15 minutes.

Conflict classification — the agent must distinguish:

- **Source conflict:** Sources disagree because one is reporting incorrectly (foreman over-reported, drone scan was partial). Expected and acceptable; surfaced gracefully. Not an agent reasoning failure.
- **Agent reasoning failure:** Agent misread or misrouted the data. Not acceptable. Triggers review and tuning.

Confidence & Clarification Flow

Confidence is always surfaced transparently to the superintendent — never hidden or averaged away. The product operates async and offline-first; clarifications are sent to foremen as non-blocking follow-up messages. Nothing waits for a clarification reply before a draft update is queued. If a reply arrives before the superintendent reviews the queue, the draft is updated in place.

Memory & State

- Agent has access to full project schedule history within a project engagement.
- Per-project trade vocabulary (zone naming conventions, terminology) is learned and retained across the project duration.
- Cross-project learning is not in scope for the pilot. Each new project initializes without prior project context.

4. Discovery & Prototyping Approach

Discovery Partners

- Existing Terabase solar customers who are building data centers
- Site superintendents from any construction field
- Specialty foremen from any industry

Prototyping Approach

Segmented prototyping follows a **Question → Prototype → Sampling → Results** flow. Three principles govern how prototypes are scoped and run:

- **One question, multiple prototypes.** A single open question can be explored through more than one prototype form — a clickable mockup and a Wizard-of-Oz simulation can run in parallel against the same question. The goal is to find the fastest signal, not to commit to one method.
- **Multiple questions, independent prototypes.** Different open questions run on separate prototype tracks simultaneously. Prototypes do not need to be unified into a single artifact before learning begins. Each question gets the smallest thing needed to answer it.
- **Sampling does not require real users.** Sampling can be conducted with actual users, but simulated use and modeled outcomes are valid — particularly when real users are inaccessible at the concept stage. Simulated sampling is explicitly not a shortcut: it is a deliberate choice about what kind of signal is needed at a given fidelity level.

Pilot-Ready Criteria

We do not advance to a live pilot until all three conditions are met:

- The buyer persona can name a specific dollar value the solution is worth to them
- The user personas can describe the workflow and articulate its value in their own words
- The internal team is operationally ready to support real-world data flows

5. Functional Requirements

Core Features

- **Offline Async Self-Reporting:** Foremen log progress via SMS or email in natural language. No app required. No active connectivity required at time of submission.
- **Schedule Evaluation Flow:** Agent reconciles field reports against P6 master schedule and generates draft updates for superintendent review.
- **Drone Scan and Update:** Reality capture ingested post-flight; agent compares against BIM and self-reports to validate or flag.

User Stories

#	Given	When	Then
1	A foreman has completed work in Zone 4	They text "Zone 4 electrical conduit done"	Agent routes to correct activity, drafts update, queues for Super review within 15 min
2	Drone and foreman reports conflict on an activity	Agent processes both inputs	Super sees both versions with conflict reason; picks one; approved version syncs to P6
3	A trade has not reported by mid-shift	Agent detects gap vs. schedule	Activity flagged; async prompt sent to foreman; Super sees gap in review queue
4	Super reviews end-of-day queue	They approve, edit, or override each draft	Approved updates sync to P6 within 15 minutes; overrides logged
5	A major site event changes the day's plan	Super makes large edits to multiple drafts	System accepts full edits without friction; no auto-approval of unchanged items

Design Principles

- **Accuracy over guesswork.** A confident wrong answer is worse than a transparent uncertain one.
- **Nail the Human-in-the-Loop moments.** The superintendent approves all P6 writes — always, no exceptions. The value of friction is real.
- **Value with a use case of 1.** Must work for one superintendent and one trade before scaling.

- **One-way reporting infrastructure.** This is not a two-way communication tool. Critical issues go through official channels. Do not create the expectation of real-time response.

Out of Scope

- Real-time or two-way communication between foremen and superintendents
- Autonomous P6 writes of any kind, regardless of confidence level
- Cross-project model learning or portfolio-level analytics (pilot phase)
- Payment, invoicing, procurement, safety, or compliance workflows

6. Evaluation & Performance Standards

Two-Axis Accuracy Model

Axis	Target	Below-Threshold Action
Task routing accuracy Right activity mapped	Near-100%. Wrong activity is never acceptable.	Immediate clarification flow to foreman. Manual review queue. Tuning required.
Magnitude accuracy % complete estimate	Best-effort with confidence score. Allowed to miss — uncertainty is surfaced.	Low-confidence estimates queued for review. Async clarification sent to reporter.

Approval Ratio Target

> **Target: 99/100** draft updates accepted with minor edits on routine days.
> Major edits on days with significant site events (weather, safety incident, scope change) are not product failures — they are expected exceptions.
> Approval ratio below 99% on routine days requires investigation and model tuning before scale.

Golden Dataset

Size	Minimum 100 labeled foreman messages across ≥5 trade types and ≥3 zone naming conventions
Owner	PM in partnership with pilot site superintendent
Validation	Domain expert labels correct activity and % complete. Agent must route correctly on ≥98 of 100 samples to pass.
Required before	Pilot launch. No pilot without a passing golden dataset run.

7. Non-Functional Requirements & Guardrails

Latency Targets

Operation	Target	Worst-Case Bound
Foreman report → superintendent review queue	**15 minutes**	30 min (degraded connectivity)
Superintendent approval → P6 sync	**15 minutes**	30 min
Drone ingest → agent suggestions available	**2 hours post-flight**	4 hours
Async clarification → foreman delivery	**15 minutes**	30 min

One-Way Design Contract

This product is reporting infrastructure, not a communication platform. Foremen post updates; the agent processes them; the superintendent reviews and acts. There is no expectation of real-time response in either direction. If a situation is time-critical, it must be escalated through official channels (radio, phone, direct conversation). The product must never create the impression that it replaces those channels.

The Escape Hatch

- Superintendent can override any agent-proposed update before or after approval.
- Approved P6 updates can be rolled back by the superintendent at any time; rollback is logged.
- If the agent cannot map a message with sufficient confidence, it surfaces the raw message to the superintendent rather than guessing silently.
- All agent actions are logged with timestamps and confidence scores.

Privacy & Data Ownership

- Job site data is owned by the GC / project owner — not Terabase.
- Default data retention: project data deleted 90 days after project close, per customer contract.
- No foreman PII stored beyond message content and timestamp. Phone numbers hashed after delivery.

8. Lifecycle & Maintenance

Degradation Thresholds

Signal	Alert Threshold	Action
Superintendent approval ratio	<99% on 3+ consecutive routine days	Investigate; pause scale; tuning sprint
Task routing accuracy	Any confirmed misroute in golden dataset re-run	Immediate tuning; do not advance pilot
P6 sync failure rate	>5% of approved updates fail to sync	Engineering escalation; manual sync fallback
Foreman report rate	<1 update/trade/day for 3+ consecutive days	PM outreach; product or adoption issue

Roadmap & Ownership

Stage	Duration	Milestone	Owner
Concept	30 days	P6 pull live; AI suggestions from drone + BIM + daily updates	PM + Engineering
Prototype	30 days	P6 update flow for Supers; SMS/email flows for Foremen	Engineering + PM
Pilot	60 days	3 customers, 3 users, 3 drone flights; contract signed	PM + Customer
Scale	90 days	First 3 co-implemented; next 3 partner-led	Partners + Product

Monitoring Cadence

Signal	Cadence	Alert Threshold
Approval ratio	Daily (pilot)	<99% on routine day
Task routing accuracy	Weekly golden dataset re-run	Any confirmed misroute
Latency (report → queue)	Continuous	>30 min average

Signal	Cadence	Alert Threshold
Foreman report rate	Daily	<1 update/trade/day
P6 sync success rate	Per sync cycle	>5% failure rate

Biggest Unknowns

- **Will foremen self-report without prompting?** The async clarification flow assumes foremen respond. If they don't, gap detection is the only fallback. Wizard-of-Oz SMS testing needed before automating the prompt layer.
- **What is the P6 write-access path?** At most hyperscale sites, P6 is owned by the GC's scheduler — not the superintendent. If updates must route through the scheduler, the approval workflow changes significantly.
- **What is the confidence threshold for queuing vs. routing?** The numeric threshold requires empirical testing against the golden dataset before it can be hardcoded.
- **What does the sales demo need to close the room?** The one moment that converts a Project Executive from interested to contracted is undefined. Must be identified during discovery before prototype stage.